ARTIFICIAL INTELLIGENCE

Lecture Set 14
Regression

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Machine Learning definition

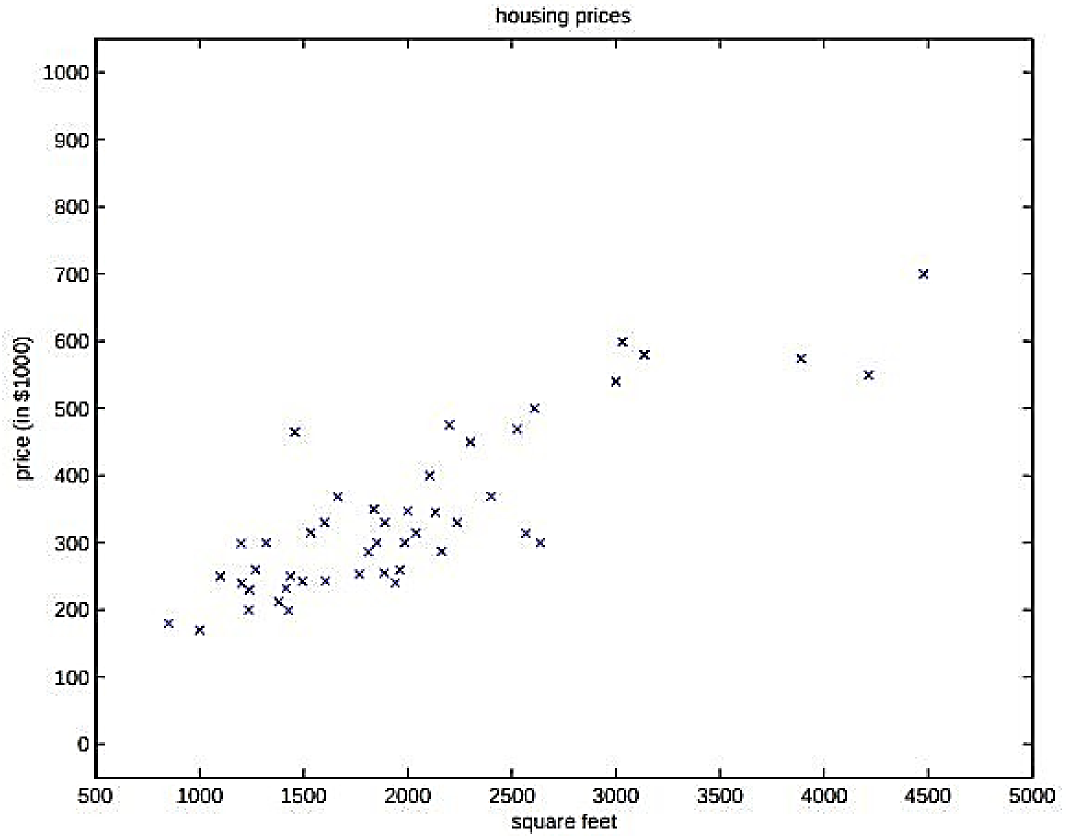
- Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.
- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to *learn* from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.

Supervised Learning

Suppose we are given a **dataset** for **supervised learning**:

Living area (feet^2)	Price (1000\$s)
2104	400
1600	330
2400	369
1416	232
3000	540
$\vdots$	$\vdots$

We can plot this data …



Decision making …

Given data like this, how can we learn to **predict** the **prices** of other houses, as a **function** of the size of their **living areas**?

Notions for variables

Input variable(s) : $\mathbf{x^{(i)}}$ **---** (living area in the previous example) --- also called input features

Output variable: $\mathbf{y^{(i)}}$ --- target variable we are trying to predict --- (price)

A pair $\mathbf{(x^{(i)}, y^{(i)})}$ is called a training example

$\mathbf{\{(x^{(i)},y^{(i)});i = 1, \ldots, m\}}$ -- a list of m training examples, training set

- **X** : space of input variables
- **Y** : space of output variables
- **X, Y** $\in$ **R** - Real number - one dimensional

Supervised learning problem

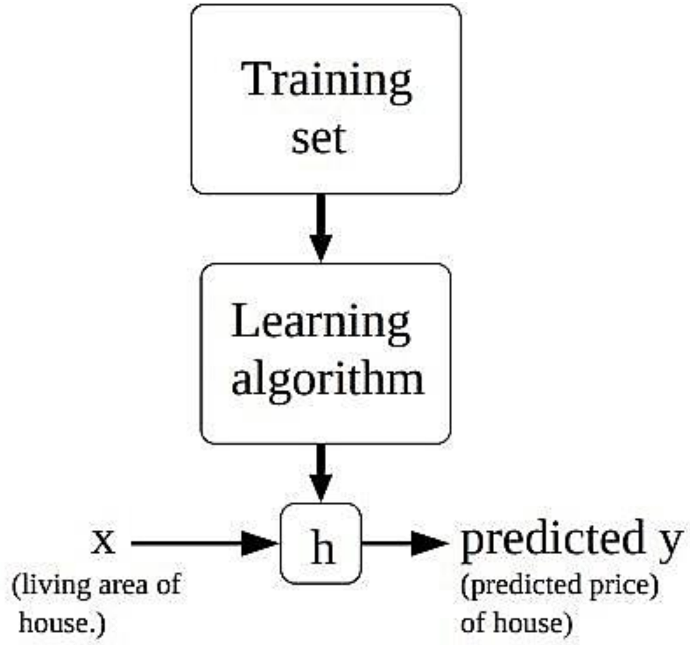
Our goal is, given a training set, to learn a function **h: X → Y**

Such that **h** is a “good predictor” of the corresponding value of y

This function **h** is called hypothesis

Different functions **h** constitute **hypotheses space of models**

Learning function h, the hypothesis



Continuous target variables

When the target variable that we are going to predict is **continuous**

As in our example of predicting **house price**

We call the (supervised) learning problem a **regression** problem

In contrast to **classification** problems, where the target variable can take on only a small number of **discrete** values.

Linear Regression

Example : Price is function of two variables or features: Living area (feet^2) and #bedrooms

x are the two-dimensional vectors in $\mathbf{R}^2$. $\mathbf{x_1^{(i)}, x_2^{(i)}}$

Living area (feet^2)	#bedrooms	Price (1000$s)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
⋮	⋮	⋮

Variable Selection, Feature Selection

When designing a learning problem, it is up to you to select variables/ features that you deem important.

The selected, devised features should aid in predicting the target variable

Representing hypothesis function

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

θ_i's are the parameters (also called weights)

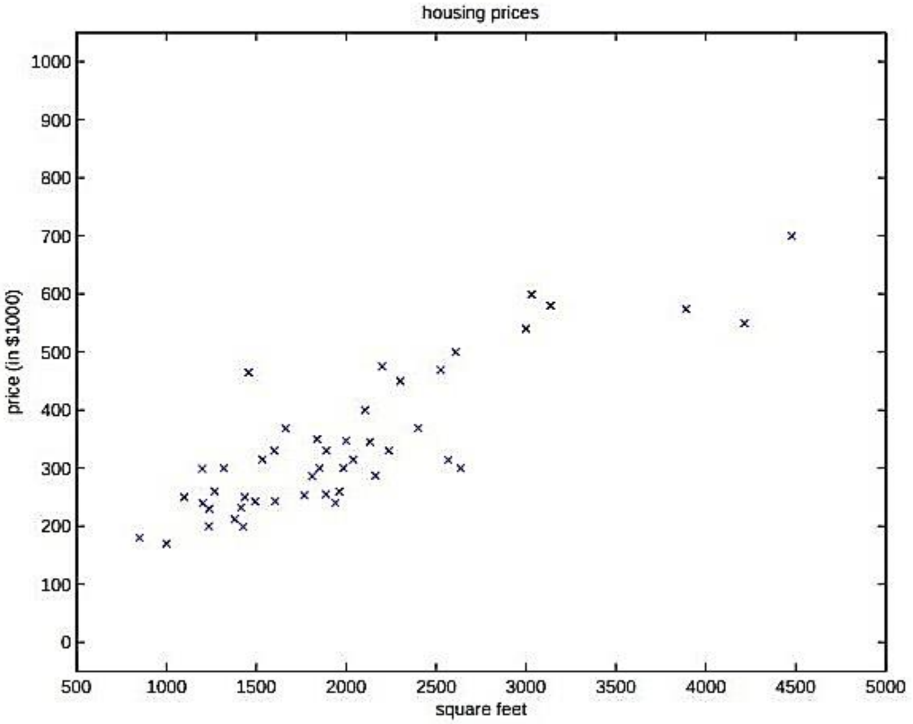
θ_i's parameterize the space of linear functions mapping from X to Y

Simplifying the notation: letting x_0 to 1 (this is the intercept term):

$$h(x) = \sum_{i=0}^{n} \theta_i x_i = \theta^T x,$$

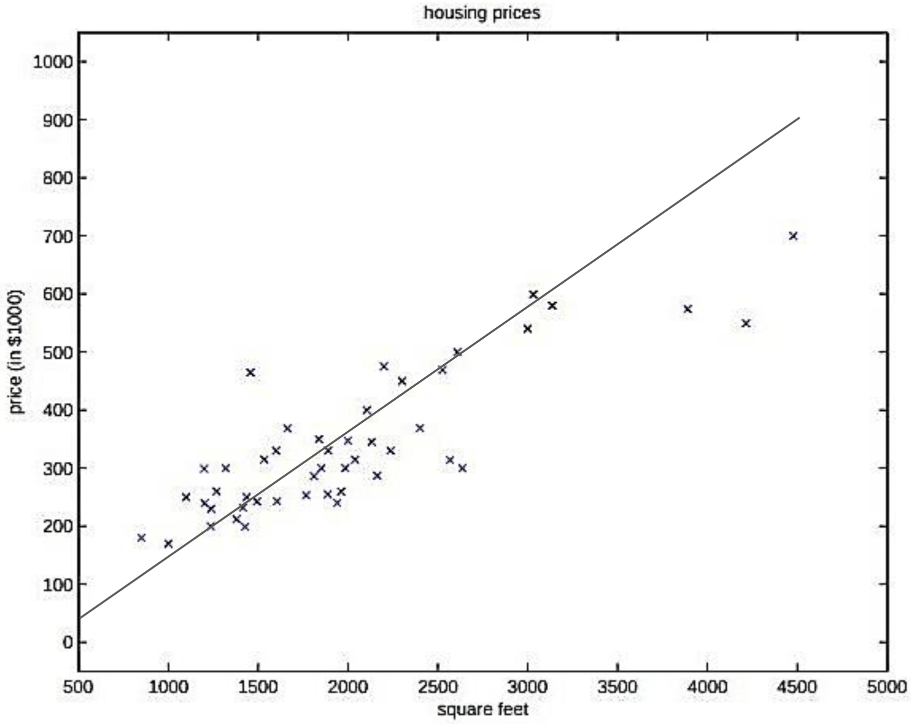
Learning the parameters

Now, given a training set, how do we pick, or learn, the parameters θ?



Learning the parameters

h(x) predictions should be close to the training set/ data



Learning the parameters

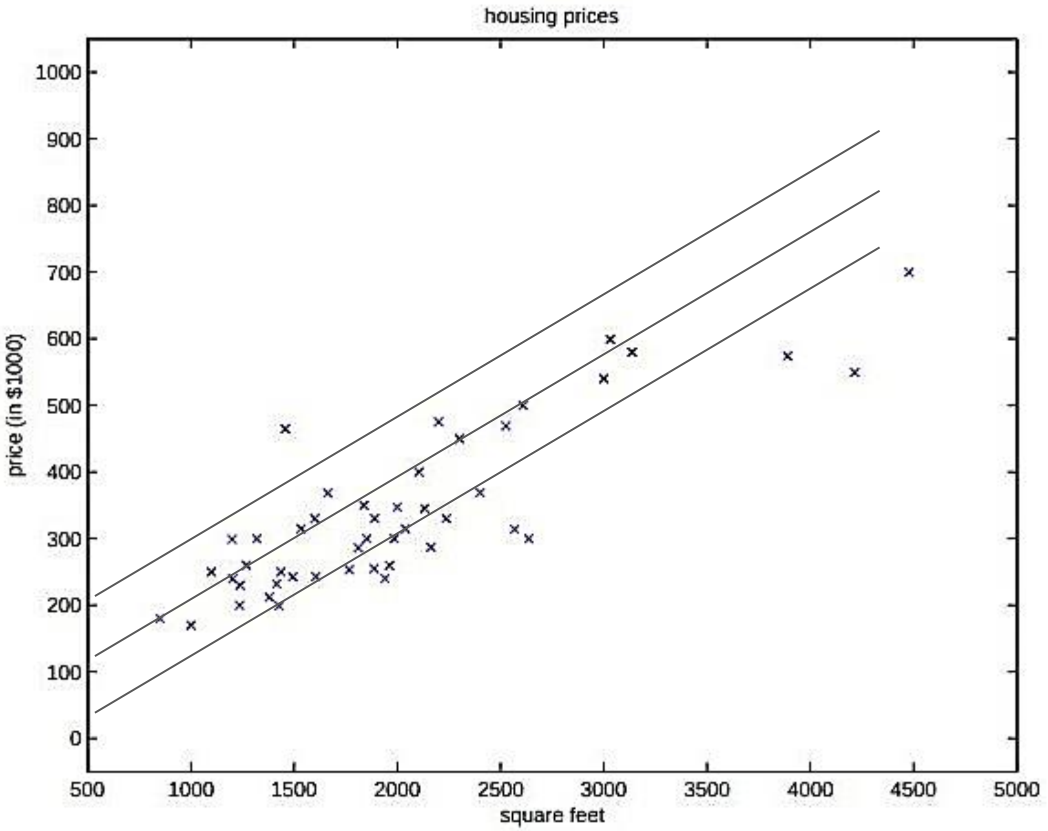
To measure how close is h(x) to y - the target variable,

We define a **cost function**, which quantify for each of θ's, how close to the h(x^i)'s are to the corresponding y^i's:

$$J(\theta) = \frac{1}{2}\sum_{i=1}^{m}(h_\theta(x^{(i)}) - y^{(i)})^2.$$

The above cost function is called **least-squares cost function**

Ordinary least squares regression model



LMS algorithm (least-mean squares)

We want to choose **θ** so as to minimize **J(θ)**

We will search for the best set values for **θ**

We start with an **initial guess of θ**, and

Repeatedly **change θ to make J(θ) smaller**

Gradient Descent Algorithm

Gradient Descent algorithm start with an initial θ, and repeatedly performs the update:

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta).$$

(This update is simultaneously performed for all values of j = 0, . . . , n.)

Here, **α** is called the learning rate

J(θ) is the partial derivative

Partial derivative of cost function (for one training example)

$$\begin{aligned}
\frac{\partial}{\partial\theta_j} J(\theta) &= \frac{\partial}{\partial\theta_j} \frac{1}{2} \left(h_\theta(x) - y\right)^2 \\
&= 2 \cdot \frac{1}{2} \left(h_\theta(x) - y\right) \cdot \frac{\partial}{\partial\theta_j} \left(h_\theta(x) - y\right) \\
&= \left(h_\theta(x) - y\right) \cdot \frac{\partial}{\partial\theta_j} \left(\sum_{i=0}^{n} \theta_i x_i - y\right) \\
&= \left(h_\theta(x) - y\right) x_j
\end{aligned}$$

Update rule: (for single training example)

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)})\right) x_j^{(i)}.$$

The rule is called the LMS update rule (LMS stands for “least mean squares”)

The magnitude of the update is proportional to the error term

Update rule: (for complete training set)

Repeat until convergence {

$$\theta_j := \theta_j + \alpha \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)}) \right) x_j^{(i)} \qquad \text{(for every } j\text{)}.$$

}

Gradient descent on original cost function J.

Computes gradient of after scanning the complete training set, called batch gradient descent

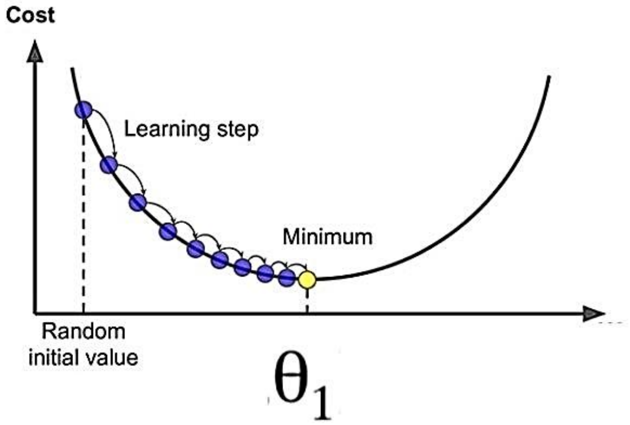
Update rule: univariate case - single parameter

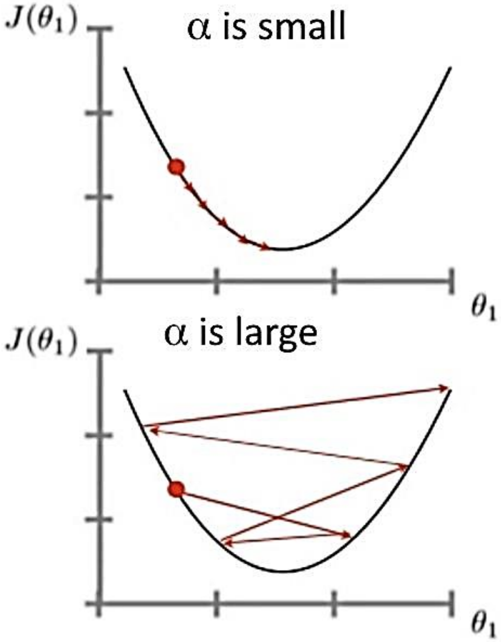
repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

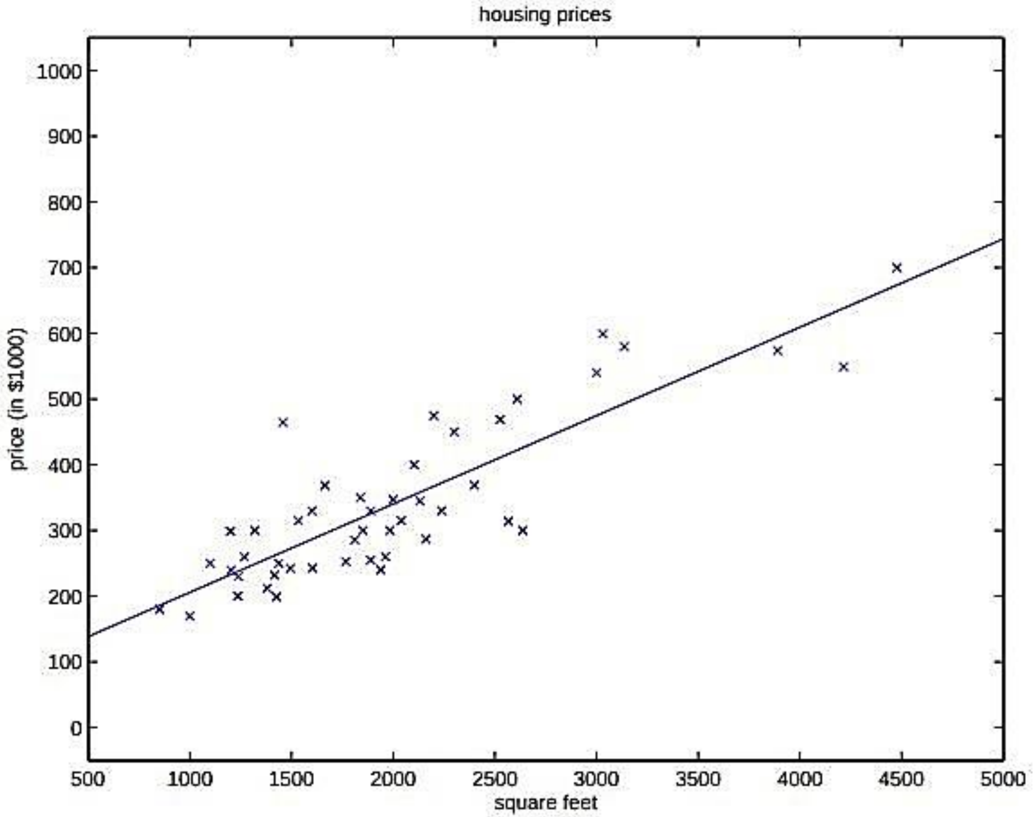
$$(\text{for } j = 1 \text{ and } j = 0)$$

}

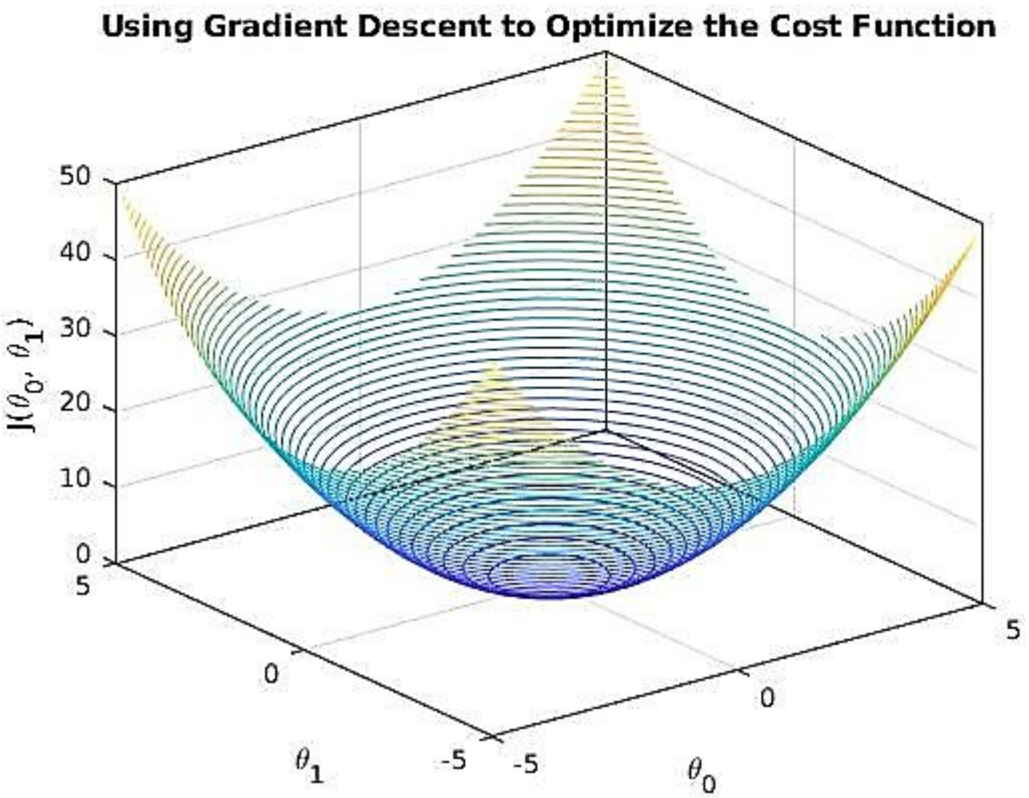




Result of LMS algorithm



Update rule: two parameters



Stochastic gradient descent

Whereas batch gradient descent has to scan through the entire training set before taking a single step—a costly operation if m is large

Stochastic gradient descent can start making progress right away, and continues to make progress with each example it looks at.

Often, stochastic gradient descent gets θ "close" to the minimum much faster than batch gradient descent.

Stochastic gradient descent

Loop {

for i=1 to m, {

$$\theta_j := \theta_j + \alpha \left(y^{(i)} - h_\theta(x^{(i)}) \right) x_j^{(i)} \qquad \text{(for every } j\text{).}$$

}

}

Linear Regression

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$

Homework

Formula for calculating slope and intercept

$$\hat{\beta} = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$\hat{\alpha} = \overline{y} - \hat{\beta}\,\overline{x}\ ,$$

- Predict the price of pizza when its diameter is 20 inches.

Diameter(X) In Inches	Price(Y) In Dollars
8	10
10	13
12	16

X	Y
1	1
2	2
3	1.3
4	3.75
5	2.25

- Let us consider an example where the five weeks' sales data (in Thousands) is given as shown in Table.
- Apply linear regression technique to predict the 7th and 12th week sales.

x_i (Week)	y_j (Sales in Thousands)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

x1 Product 1 Sales	x2 Product 2 Sales	Y Weekly Sales
1	4	1
2	5	6
3	8	8
4	2	12

$Y = 2.746X_1 - 1.52X_2 + 0.212$

$Y = 0.425X + 0.785$

X	Y	XY	X^2
1	1	1	1
2	2	4	4
3	1.3	3.9	9
4	3.75	15	16
5	2.25	11.25	25
15	10.3	35.15	55

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
3	8	-3.7	9	64	24	-11.1	-29.6
4	5	3.5	16	25	20	14	17.5
5	7	2.5	25	49	35	12.5	17.5
6	3	11.5	36	9	18	69	34.5
2	1	5.7	4	1	2	11.4	5.7
20	24	19.5	90	148	99	95.8	45.6

$$m = \frac{n\sum XY - \sum X \sum Y}{n(\sum X^2) - (\sum X)^2}$$

$$m = \frac{5(35.15) - (15)(10.3)}{5(55) - (15)^2}$$

$$m = \frac{21.25}{50} = 0.425$$

$$c = \bar{Y} - m\bar{X}$$

$$c = \frac{10.3}{5} - (0.425)\left(\frac{15}{5}\right) = 0.785$$

$$b_1 = \frac{(\sum X_2^2)(\sum X_1Y) - (\sum X_1X_2)(\sum X_2Y)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1X_2)^2} = \frac{(148)(95.8) - (99)(45.6)}{(90)(148) - (99)^2}$$

$$b_1 = 2.746$$

$$b_2 = \frac{(\sum X_1^2)(\sum X_2Y) - (\sum X_1X_2)(\sum X_1Y)}{(\sum X_1^2)(\sum X_2^2) - (\sum X_1X_2)^2} = \frac{(90)(45.6) - (99)(95.8)}{(90)(148) - (99)^2}$$

$$b_2 = -1.52$$

$$a = \bar{Y} - b_1\bar{X}_1 - b_2\bar{X}_2 = \frac{19.5}{5} - (2.746)\left(\frac{20}{5}\right) + (1.52)\left(\frac{24}{5}\right) = 0.212$$

Linear Regression 2 independent variable

SUBJECT	Y	X_1	X_2	X_1X_1	X_2X_2	X_1X_2	X_1Y	X_2Y
1	-3.7	3	8	9	64	24	-11.1	-29.6
2	3.5	4	5	16	25	20	14	17.5
3	2.5	5	7	25	49	35	12.5	17.5
4	11.5	6	3	36	9	18	69	34.5
5	5.7	2	1	4	1	2	11.4	5.7
Σ	19.5	20	24	90	148	99	95.8	45.6

$$a = \overline{Y} - b_1\overline{X}_1 - b_2\overline{X}_2$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1y) - (\sum x_1x_2)(\sum x_2y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1x_2)^2}$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2y) - (\sum x_1x_2)(\sum x_1y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1x_2)^2}$$

$$b_1 = \frac{(\sum x_2^2)(\sum x_1y) - (\sum x_1x_2)(\sum x_2y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1x_2)^2} = \frac{32.8 * 17.8 - 3 * (-48)}{10 * 32.8 - 3 * 3} = 2.28$$

$$b_2 = \frac{(\sum x_1^2)(\sum x_2y) - (\sum x_1x_2)(\sum x_1y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1x_2)^2} = \frac{10 * (-48) - 3 * 17.8}{10 * 32.8 - 3 * 3} = -1.67$$

$$a = \overline{Y} - b_1\overline{X}_1 - b_2\overline{X}_2 = \frac{19.5}{5} - \frac{2.28 * 20}{5} - \frac{-1.67 * 24}{5} = 2.796$$

Final Regression equation or Model is:

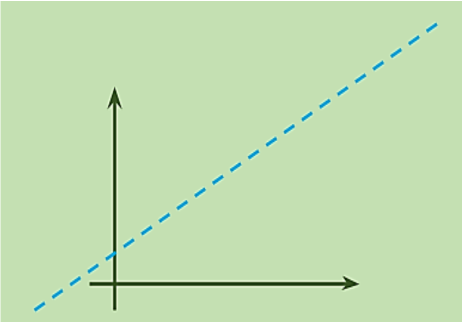
$$\boldsymbol{Y = 2.796 + 2.28\, x_1 - 1.67 x_2}$$

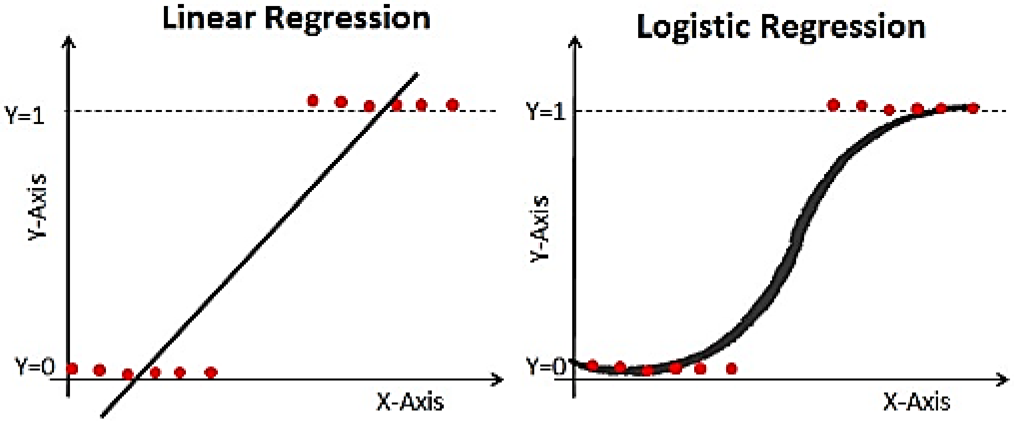
$$Now\ given \quad x_1 = 3 \quad and \quad x_2 = 2 \qquad Y = ?$$

$$Y = 2.796 + 2.28 * 3 - 1.67 * 2$$
$$= \mathbf{6.296}$$

Problem in Linear Regression

- Not all data is linear
- As x increases y tends to approach infinity.
- Not the case in reality.





$$h_\theta(x) = \theta_0 + \theta_1 x_1$$

$$p = \frac{1}{1+e^{-z}}$$

Sigmoid function brings non-linearity. Based on a threshold value, classes can be classified using the probability value we get after applying sigmoid function.

Logistic Regression – Solved Example 1

- The dataset of pass or fail in an exam of 5 students is given in the table.
- Use logistic regression as classifier to answer the following questions.

1. Calculate the probability of pass for the student who studied 33 hours.

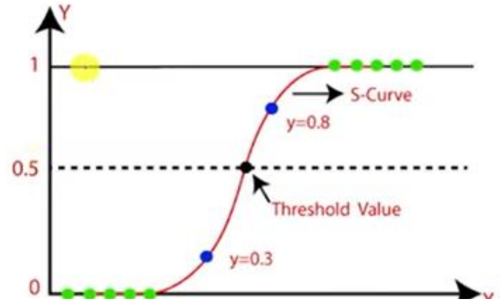
Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- We use Sigmoid Function in logistic regression

Assume the model suggested by the optimizer for odds of passing the course is,

$$\log(odds) = -64 + 2 * hours$$

- $s(x) = \frac{1}{1+e^{-x}}$



Logistic Regression – Solved Example 1

1. Calculate the probability of pass for the student who studied 33 hours.

$$s(x) = \frac{1}{1 + e^{-x}}$$

- $p = \frac{1}{1+e^{-z}}$
- $z = -64 + 2 * 33 = -64 + 66 = 2$
- $p = \frac{1}{1+e^{-2}} = 0.88$
- That is, if student studies 33 hours, then there is **88% chance** that the student will pass the exam

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$\log(odds) = z = -64 + 2 * hours$$

https://www.youtube.com/watch?v=2C8IqOLO1os